

A top-down view of a grey desk with various items: a silver stethoscope, a laptop displaying a medical website, a pair of glasses, a black smartphone, a white coffee cup, and a spiral-bound notebook with a pen. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

MENINGITIS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

BBB	Blood-Brain Barrier
CNS	Central Nervous System
CSF	Cerebrospinal Fluid
ED	Emergency Department
HIC	High-income country
HIV	Human Immunodeficiency Virus
HSV	Herpes Simplex Virus
IV	Intravenous
NSAIDs	Non-steroid Anti-inflammatory Drugs
PO	By Mouth
Q	Every
SMX-TMP	Sulfamethoxazole-Trimethoprim
WBC	White Blood Cells

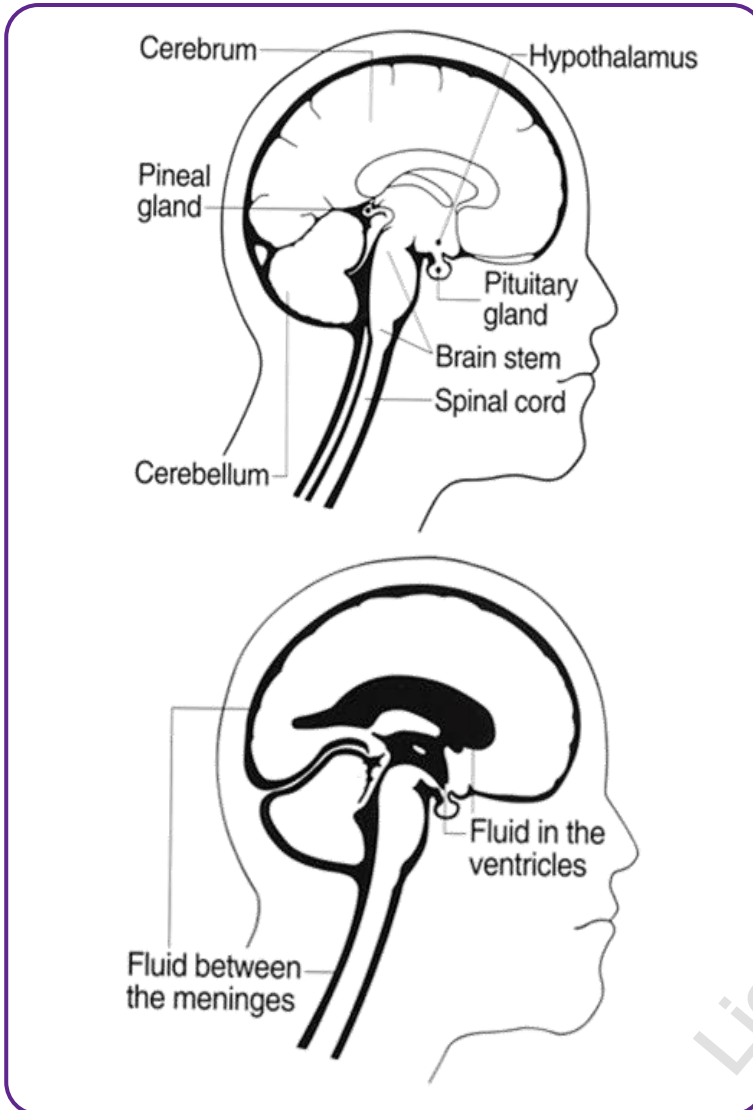
CASE #1

FJ is a 68-year-old woman with a history of type 2 diabetes and chronic sinusitis. She presents with sudden-onset fever, headache, and altered mental status. Her family reports she became increasingly confused and lethargic over the past 12 hours. On examination, she is febrile, disoriented, and has neck stiffness with a positive Kernig sign. A CT head is performed and shows no contraindications to lumbar puncture. CSF analysis reveals elevated opening pressure, neutrophilic pleocytosis, low glucose levels, and elevated protein levels.

What is the appropriate therapy given the available information?

WHAT IS MENINGITIS?

FYI



Inflammation of the meninges due to a virus, bacteria, fungus, or aseptic source (e.g. medication-induced)

1. CSF is secreted from the ventricles of the brain

2. Meninges contain CSF

Cerebrospinal fluid (CSF) circulation process

4. CSF circulates between the brain and spine

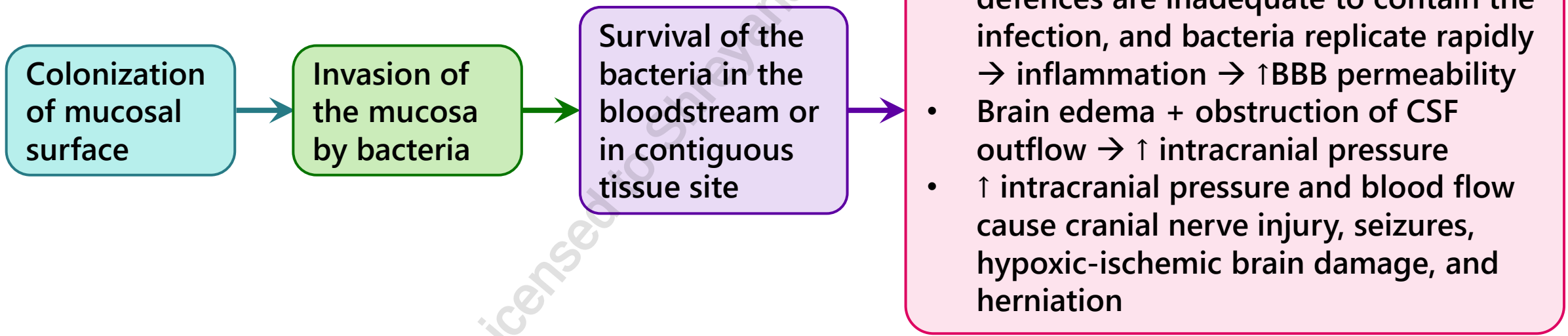
3. CSF moves towards the spinal cord

PATHOGENESIS- STEPS TO INFECTION

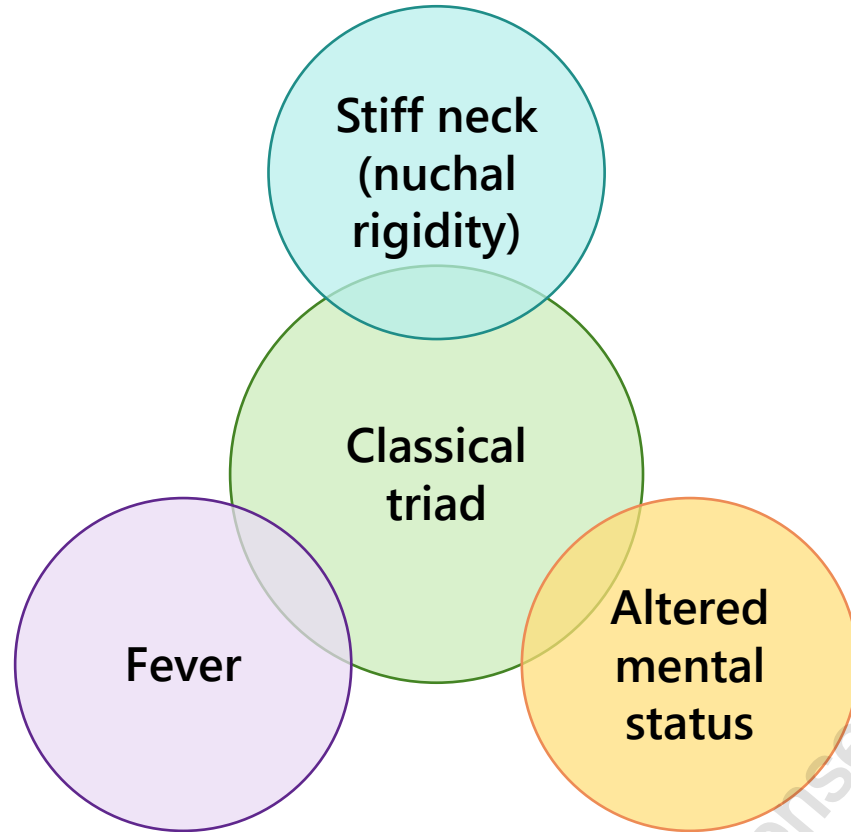
FYI

Mechanisms of infection:

- Hematogenous spread (most common)
- Contiguous spread: from a para-meningeal focus (sinusitis, otitis media)
- Direct inoculation: head trauma, neurosurgery



CLINICAL PRESENTATION



Other symptoms

- Anorexia
- Nausea, vomiting
- Irritability
- High respiratory rate
- Low blood pressure
- Headache
- Petechial rash
- Photophobia
- Seizures
- Septic shock
- Focal neurological deficit
- Decreased Glasgow coma scale
- Brudzinski sign: reflex flexion of hips and knees produced on flexion of neck when lying recumbent
- Kernig's sign: pain when extending hamstrings when lying supine with thighs perpendicular to trunk

RISK FACTORS

- Age:
 - Infants < 1 month of age
 - Children < 4 years of age
 - Elderly
- Close contact environments (school, military barracks, prison)
- Exposure to patients colonized or infected with *N. meningitidis* or *H. influenzae*
- Neurosurgery/head trauma
- Immunocompromised patients (e.g. alcohol use disorder, HIV, chemotherapy)
- Sinusitis, otitis media

COMPLICATIONS OF BACTERIAL MENINGITIS

FYI

- Neurologic sequelae develop in $\frac{1}{3}$ - $\frac{1}{2}$ of patients
- Spastic paraparesis, behavioural disorders, learning deficits
- Inappropriate secretion of anti-diuretic hormone → hyponatremia and worsening cerebral edema
- Children: pneumococcal meningitis highest risk of permanent neurological sequelae
- Neurologic sequelae especially sensorineural hearing loss:
S. pneumoniae > *N. meningitidis* > *H. influenzae*

DIAGNOSTIC TESTS AND DIFFERENTIAL DIAGNOSIS

Laboratory tests to diagnose bacterial meningitis

- Serum sodium: Low
- CSF culture
- CSF PCR
- Latex particle agglutination
 - Detects antigens of *bacteria and/or fungi*

Differential diagnosis

- Fungal meningitis: CSF may appear clear
- Viral meningitis: commonly caused by enterovirus, possibly HSV; CSF may appear clear
- Aseptic meningitis: drug causes include murine monoclonal antibodies, SMX-TMP, NSAIDs, azathioprine

GOALS OF THERAPY

- ✓ Cure bacterial meningitis infection by using high dose IV antibiotic
- ✓ Eliminate symptoms and normalize blood work
- ✓ Prevent complications
- ✓ Prevent adverse drug reactions from drug therapy

PRINCIPLES OF DRUG THERAPY

Meningitis is life-threatening

Immediately start empiric IV antibiotics + dexamethasone after lumbar puncture is done or if delayed lumbar puncture

Cover all suspected organisms and consider adding drugs to overcome resistance

Use bactericidal drugs with CNS penetration and optimize dose (high dose)

EMPIRIC DRUG THERAPY

Age group	Most likely organisms	Recommended empiric therapy
Neonates: < 1 month	• Group B streptococcus (<i>Streptococcus agalactiae</i>) • <i>E. coli</i>, and other gram-negative bacilli (<i>Klebsiella</i>, <i>Serratia</i> species) • <i>Listeria monocytogenes</i>	• Ampicillin + cefotaxime OR • Ampicillin + aminoglycoside
Infants and children: ≥ 1 month – 23 months	• <i>Streptococcus pneumoniae</i>, <i>N. meningitidis</i>, <i>S. agalactiae</i>, <i>H. influenzae</i>, <i>E. coli</i>	• Cefotaxime + vancomycin OR • Ceftriaxone + vancomycin
Older children and adults: 2-50 years	• <i>N. meningitidis</i>, <i>S. pneumoniae</i>, <i>H. influenzae</i>	
>50 years old	• <i>S. pneumoniae</i>, <i>N. meningitidis</i>, <i>L. monocytogenes</i>, gram-negative bacilli	Ampicillin + (cefotaxime OR ceftriaxone) + vancomycin
Immunocompromised: Alcohol use disorder, altered immune status, pregnancy		

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What pathogen is most likely causing this patient's meningitis?

EMPIRIC DRUG THERAPY (FOR THOSE WITH SERIOUS BETA-LACTAM ALLERGIES)

Age group	Most likely organisms	Recommended empiric therapy
Infants and children: ≥ 1 month – 23 months	• <i>Streptococcus pneumoniae</i>, <i>N. meningitidis</i>, <i>S. agalactiae</i>, <i>H. influenzae</i>, <i>E. coli</i>	• Meropenem + vancomycin OR • Chloramphenicol + vancomycin • (add TMP/SMX if immunocompromised)
Older children and adults: 2-50 years	• <i>N. meningitidis</i>, <i>S. pneumoniae</i>, <i>H. influenzae</i>	
> 50 years old	• <i>S. pneumoniae</i>, <i>N. meningitidis</i>, <i>L. monocytogenes</i>, gram-negative bacilli	TMP/SMX+ meropenem + vancomycin TMP/SMX + moxifloxacin + vancomycin
Immunocompromised: Alcohol use disorder, altered immune status, pregnancy		

TARGETED DRUG THERAPY- DRUG THERAPY AFTER OBTAINING CULTURE RESULTS

Pathogen	First-line therapy	Days
<i>H. influenzae</i>	• Beta-lactamase-negative: IV ampicillin • Beta-lactamase-positive: 3rd generation cephalosporin (cefotaxime or ceftriaxone)	7-10
<i>N. meningitidis</i>	If susceptible: penicillin G OR 3 rd generation cephalosporin (cefotaxime or ceftriaxone)	7
<i>S. pneumoniae</i>	• Penicillin G OR 3rd generation cephalosporin (cefotaxime or ceftriaxone) +/- vancomycin • Combination and drug selections depend on susceptibility	10-14
<i>S. agalactiae</i>	Penicillin G or ampicillin + gentamycin for synergy	14-21
<i>L. monocytogenes</i>		≥ 21
Gram-negative bacilli	3 rd generation cephalosporin (cefotaxime or ceftriaxone) +/- aminoglycoside (gentamycin or tobramycin) or meropenem or imipenem/cilastatin	21

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What is the appropriate therapy given the available information?

TARGETED DRUG THERAPY- DRUG THERAPY AFTER OBTAINING CULTURE RESULTS

FYI

Pathogen	Alternative therapy/severe beta-lactam allergy (choose one)	Days
<i>H. influenzae</i>	Ciprofloxacin/Levofloxacin/Moxifloxacin Meropenem Chloramphenicol	7-10
<i>N. meningitidis</i>	Ciprofloxacin/Levofloxacin/Moxifloxacin Meropenem Chloramphenicol	7
<i>S. pneumoniae</i>	Vancomycin+ Rifampin Meropenem Levofloxacin/Moxifloxacin Chloramphenicol	10-14
<i>L. monocytogenes</i>	TMP-SMX Linezolid	≥ 21
Gram-negative bacilli	Meropenem Ciprofloxacin/Levofloxacin/Moxifloxacin Chloramphenicol	21

ADJUNCTIVE TREATMENT- DEXAMETHASONE

- Reduces cerebral edema
- Lowers intracranial pressure
- Steroids ↓ the inflammation which causes neurologic complications
- Most beneficial for prevention of hearing loss and short term neurologic sequelae in HICs
- Data supportive of therapy in adults for *S. pneumoniae*
- Data supportive of therapy in children for *H. influenzae*
- Most beneficial when given before or at the time of antibiotic administration
- Dosing in adults: dexamethasone 10 mg IV every 6 hours for 4 days should be given before or at the same time as the first dose of parenteral antibiotic
- Dosing in children: 0.15 mg/kg IV every 6 hours for 2-4 days

PROPHYLAXIS

Chemoprophylaxis	Vaccination prophylaxis
• Only recommended to decrease risk of oropharyngeal carriage in certain bacteria (<i>N. meningitidis</i> and <i>H. influenzae</i>) in close contact individuals • Exposure to <i>N. meningitidis</i>: • rifampin OR • ciprofloxacin OR • Pregnancy: ceftriaxone • Exposure to <i>H. influenzae</i>: • Rifampin OR • Pregnancy: ceftriaxone	• Vaccination with quadrivalent meningococcal vaccine (A/C/Y/W-135) +/- (B): • Over 90% effective in children > 2 years • Vaccination with <i>H. influenzae type b</i> vaccine: • Over 95% effective • Most likely the reason why <i>H. influenzae</i> meningitis not seen as often in Canada

MONITORING PARAMETERS

Parameter	Degree of change	Timeframe
Efficacy endpoints		
Symptom improvement (e.g. headache, fever)	Return to normal	1-2 days
Blood and CSF Chemistry	Return to normal	1 day
Mental status and vitals (e.g. tachypnea)	Return to normal	1-2 days
Safety endpoints		
Drug side effects (e.g. flushing with vancomycin)	Minimize and manage	Ongoing during treatment

TAKEAWAY

- Meningitis = Inflammation of the meninges due to a virus, bacteria, fungus, or aseptic source (e.g. medication-induced)
- Clinical symptoms: stiff neck, altered mental status, fever
- **Empiric drug therapy** for bacterial meningitis:
 - < 1 month: **ampicillin + cefotaxime** or **ampicillin + aminoglycoside**
 - 1 month to 50 years of age: **vancomycin + 3rd generation cephalosporin** (ceftriaxone or cefotaxime)
 - > 50 years of age: **vancomycin + ampicillin + 3rd generation cephalosporin** (ceftriaxone or cefotaxime)
 - Immunocompromised (alcohol use disorder, altered immune status, pregnancy): **vancomycin + ampicillin + 3rd generation cephalosporin** (ceftriaxone or cefotaxime)
- **Dexamethasone:**
 - Reduces cerebral edema and lowers intracranial pressure
 - Best evidence for use in adults for *S. pneumoniae* and in children for *H. influenzae*
 - Most beneficial when given before or at the time of antibiotic administration

CASE 1

A 19-year-old patient, JM, is brought to the ED by friends for fever and confusion. JM, a student enrolled at a local university, has been studying for exams for the last week. Beginning the night before presentation to the ED, he complained of a headache and some light sensitivity and went to bed. The following morning, he did not come to breakfast and his friends found him in bed, confused and warm to the touch. JM has a past medical history of rheumatoid arthritis for which he takes infliximab SC qmonth and naproxen 500 mg PO Q8H PRN. He has an allergy to penicillin (anaphylaxis age 16). He is a student and drinks alcohol regularly and uses cannabis occasionally.

1. What is not an important principle of choosing antibiotic therapy?
 - a) Broad spectrum antimicrobial therapy
 - b) Good oral bioavailability
 - c) Good CNS penetration
 - d) Maximizing antibiotic doses



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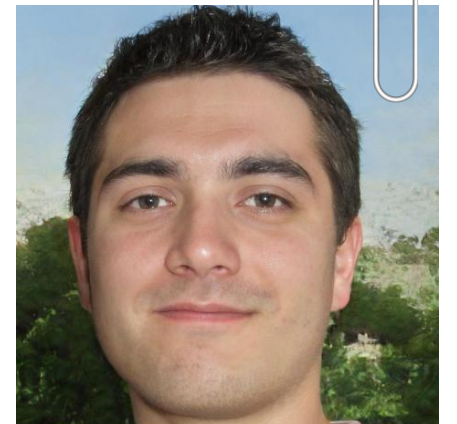
2. What is the BEST choice for initial empiric antimicrobial therapy?
- a) Ampicillin + moxifloxacin + vancomycin
 - b) Ampicillin + ceftriaxone + vancomycin
 - c) TMP/SMX + vancomycin
 - d) TMP/SMX + vancomycin + meropenem



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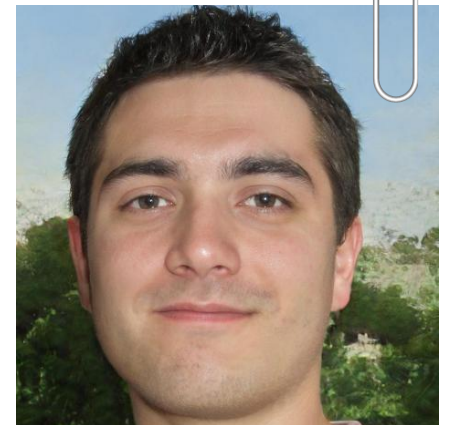
3. His multiplex PCR returns with *S. pneumoniae* on day 1 of admission, what is the optimal decision for antimicrobial therapy at this point?
- a) Vancomycin
 - b) Ceftriaxone + linezolid
 - c) Vancomycin + rifampin
 - d) Chloramphenicol



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4. On day 2 of hospital admission, the culture and susceptibility return for his isolate. Assuming it is susceptible to all tested therapy, what is the optimal choice and duration of targeted antimicrobial treatment assuming no complications?
- a) Vancomycin x 8 days
 - b) Moxifloxacin x 12 days
 - c) TMP/SMX x 8 days
 - d) Ertapenem x 12 days



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CHANGE LOG

March 2025

- Slide 9: Diagnosis updated
- Slide 15: Updated drug options

May 2025

- Slide 16: Changed dexamethasone duration in children from 4 days to 2-4 days

July 2025

- Formatting changes made throughout (some slides rearranged)
- Slides 4,14,15,18: Case study added
- Slides 13 and 16: Changed age range to match IDSA guidelines

August 2025

- Slides 8, 13, 16: Changed 'alcoholism' to 'alcohol use disorder'
- Slide 10: Removed 'FYI'

October 2025

- No content changes made

February 2026

- Slide 2: Updated NAPRA competencies
- Slide 23: Added takeaways